

Project I: Geometric Measurement for MiC Quality Inspection

🌸 Task I: Geometric Measurement of MiC Module for MiC Quality Inspection --- Dr. Hong Jie

★ MiC?

- ★ MiC: an advanced off-site construction method that involves manufacturing entire volumetric modules in a factory before transporting and assembling them on site
- ★ Quality Assessment of MiC: comprehensive measurements of characteristics and parts

★ Typical Measurement Items

- ★ **MiC** Length/width/height
- ★ **Door/windows** size
- ★ **Wall/ground** roughness
- ★ **Pull hole** size/center position
- ★ **Embedded parts** center position/height

MiC
length/width/height



Door/windows



Embedded part



MiC Assembly
(from Saint-Gobain Weber HK)

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✿ Dataset

- ✿ Raw data collected: 2 point clouds of 2 MiC modules scanned by 3D scanner.
- ✿ The measurement results by human work as the ground truth.



Collected 2 Point Clouds of 2 MiC modules in Factories

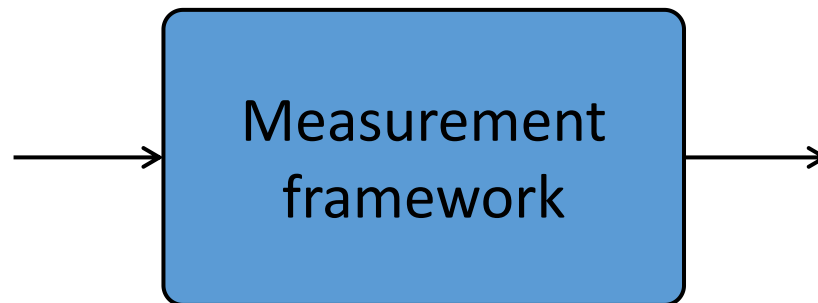
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✿ Task Objectives

- ✿ Propose/Design the measurement framework
- ✿ Achieve the measurement results



Input point cloud



- 1) Room size: length/width/height
- 2) Windows/door size: width/height
- 3) Wall roughness

Measurement results

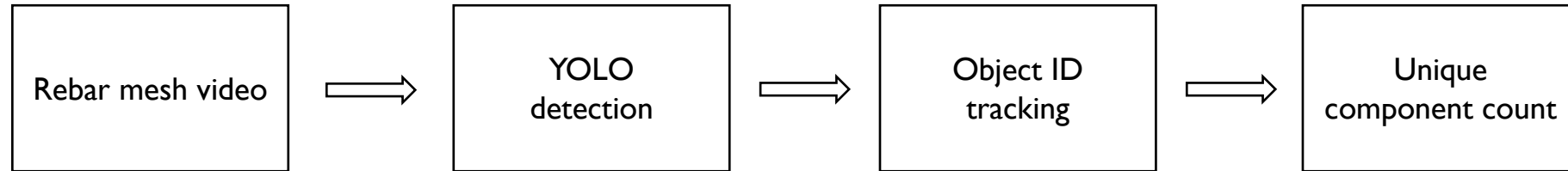
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✿ Task Evaluations

- ✿ Comprehensive literature review on geometric measurement
- ✿ The novelty of the proposed measurement framework
- ✿ Achieve the reasonable measurement performances
 - ★ Measurement error

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✿ Task 2: Embedded component detection and counting with YOLO --- Dr. LI Tingtian



✿ Task Objectives

- Use our provided images of embedded components and annotate bounding boxes and classes. Train or fine-tune a YOLO detector.
- Run YOLO on the rebar-mesh video. Track each detected component across frames and count unique IDs to avoid duplicate counting.
- Write a report including methodology and implementation details, quantitative and qualitative demonstration of the results, analysis of your results.

